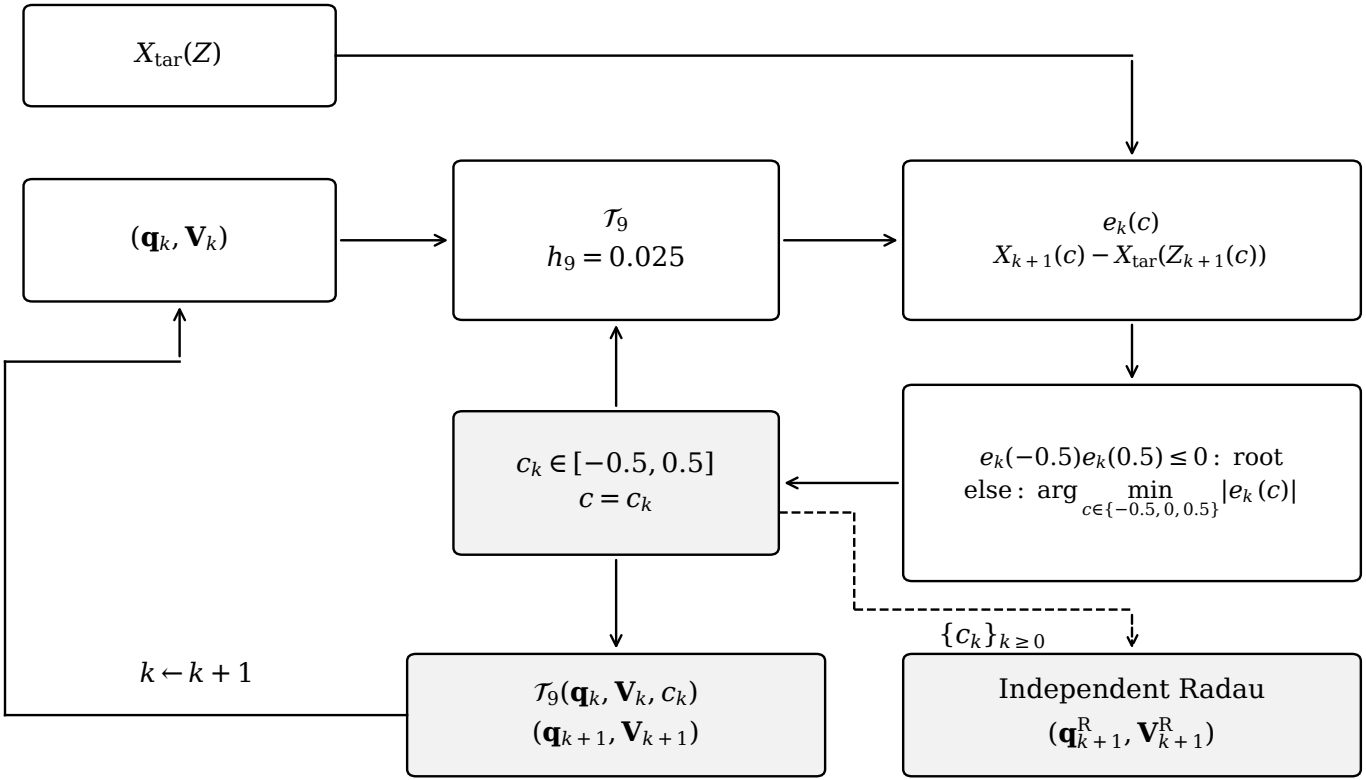
$X_{\mathrm{tar}}(Z)$
$(\mathbf{q}_k, \mathbf{V}_k)$
$\mathcal{T}_9$
$h_9 = 0.025$
$e_k(c)$
$X_{k+1}(c) - X_{\mathrm{tar}}(Z_{k+1}(c))$
$c_k \in [-0.5, 0.5]$
$c = c_k$
$e_k(-0.5)e_k(0.5) \le 0$: root
else : $\arg\min_{c \in \{-0.5, 0, 0.5\}} |e_k(c)|$
$\{c_k\}_{k \ge 0}$
$k \leftarrow k+1$
$\mathcal{T}_9(\mathbf{q}_k, \mathbf{V}_k, c_k)$
$(\mathbf{q}_{k+1}, \mathbf{V}_{k+1})$
Independent Radau
$(\mathbf{q}^{\mathrm{R}}_{k+1}, \mathbf{V}^{\mathrm{R}}_{k+1})$